Prepared for submission to JCAP

Reconstructing the primordial power spectrum with FrediPy: a proof of concept

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Abstract. The CMB temperature spectrum is a linear integral transform of the primordial curvature spectrum when the background cosmology and transfer functions are fixed. This makes it a natural test case for FrediPy, a general Python package for Bayesian inversion of Fredholm integral equations. We formulate the CMB projection as a Gaussian-process inverse problem and retain only the ingredients needed for a transparent proof of concept. First, we reconstruct a known power law from synthetic, unlensed CLASS temperature data. On the fixed range $3 \times 10^{-4} < k < 0.15\,\mathrm{Mpc}^{-1}$, the posterior mean differs from the input by at most 7.5×10^{-4} fractionally. We then apply the same public FrediPy interface to Planck PR3 high-ℓ TT bandpowers, keeping the cosmology, lensing template and calibration fixed while varying the Gaussian-process hyperparameters in a small sensitivity scan. At the reference values and on the range with direct noise-weighted operator sensitivity, the flexible posterior mean differs from its best-fitting power law by at most 1.24% and 0.48 pointwise posterior standard deviations. Their data quadratics are 218.5 and 218.9, respectively. Factor-two changes of either RBF hyperparameter keep the maximum separation below 2.9%. FrediPy therefore passes the controlled reconstruction test, while the observational example gives no reason to replace a featureless power law in this conditional setting. The latter is an illustration, not a feature-significance or cosmological-parameter analysis.

Keywords: Bayesian reasoning, CMBR theory, cosmological perturbation theory

1 Introduction

A power law provides the standard two-parameter description of primordial fluctuations,

$$\mathcal{P}_{\mathcal{R}}(k) = A_s \left(\frac{k}{k_*}\right)^{n_s - 1}. \tag{1.1}$$

It is nearly scale invariant and becomes a straight line when both axes are logarithmic. CMB observations are consistent with this form [1, 2], but they do not measure $\mathcal{P}_{\mathcal{R}}(k)$ directly. They measure angular spectra after acoustic evolution and projection along the line of sight. Reconstructing the input from the output is therefore an inverse problem.

There is a long history of approaching this problem through deconvolution, regularised direct inversion, knots and other non-parametric descriptions [3–8]. The common difficulty is physical rather than algorithmic: many primordial functions project to almost the same angular spectrum. A reconstruction must state which combinations of $\mathcal{P}_{\mathcal{R}}$ the data constrain and how the remaining freedom is regularised.

FrediPy was developed as a general solver for this class of problem [9, 10]. It places a Gaussian-process prior on an unknown function and conditions that process on noisy linear integral constraints. Although its initial applications concerned spectral functions, neither its mathematics nor its software interface is tied to that setting. The primordial spectrum is a particularly clean cosmological example because the unlensed CMB spectra are Fredholm integrals of the first kind at fixed cosmology.

Our contribution is deliberately narrow: we do not introduce another primordial-spectrum reconstruction algorithm. Instead, we supply the CMB-specific operator, data vector and covariance to the unmodified public FrediPy interface. This provides an inspectable test of whether a general Fredholm solver developed for QCD spectral functions can both recover a CLASS input and condition on Planck bandpowers.

Here we ask whether FrediPy can recover a known primordial spectrum from CLASS and whether the same construction can be shown on real CMB bandpowers without building a full cosmological inference pipeline around it. The answer to the first question is quantitative: the input power law is recovered to better than one part in a thousand over the tested range. For the second, a conditional Planck TT reconstruction remains close to an optimised power law. This does not test inflationary models or establish the absence of features. It shows what the Fredholm formulation does and where a larger analysis would have to begin.

2 The CMB projection as a Fredholm problem

2.1 Forward map

We use the dimensionless curvature spectrum defined by

$$\langle \mathcal{R}(\boldsymbol{k})\mathcal{R}(\boldsymbol{k}')\rangle = (2\pi)^3\delta^{(3)}(\boldsymbol{k}+\boldsymbol{k}')\,\frac{2\pi^2}{k^3}\mathcal{P}_{\mathcal{R}}(k). \tag{2.1}$$

For fixed late-time parameters and unlensed scalar perturbations, the temperature spectrum is

$$C_\ell^{TT} = 4\pi\int \mathrm{d}\ln k\,\mathcal{P}_{\mathcal{R}}(k)\left|\Delta_\ell^T(k)\right|^2, \tag{2.2}$$

where Δ_ℓ^T is the temperature transfer function. CLASS computes the transfer functions and the corresponding angular spectra [11]. Equation (2.2) has the standard Fredholm form

$$g(x) = \int \mathrm{d}t\,G(x,t)f(t). \tag{2.3}$$

Here x labels a multipole or bandpower, $t = \ln k$, the unknown is $f = \mathcal{P}_{\mathcal{R}}$, and the kernel contains the squared radiation transfer function.

The kernel is broad and oscillatory. Projection mixes neighbouring wavenumbers, while Silk damping and the finite observed multipole range remove information. Inverting equation (2.2) without regularisation would amplify small numerical or observational fluctuations. Figure 1 visualises the discretised response. It also makes clear why a CMB experiment constrains a range of k, rather than each node independently.

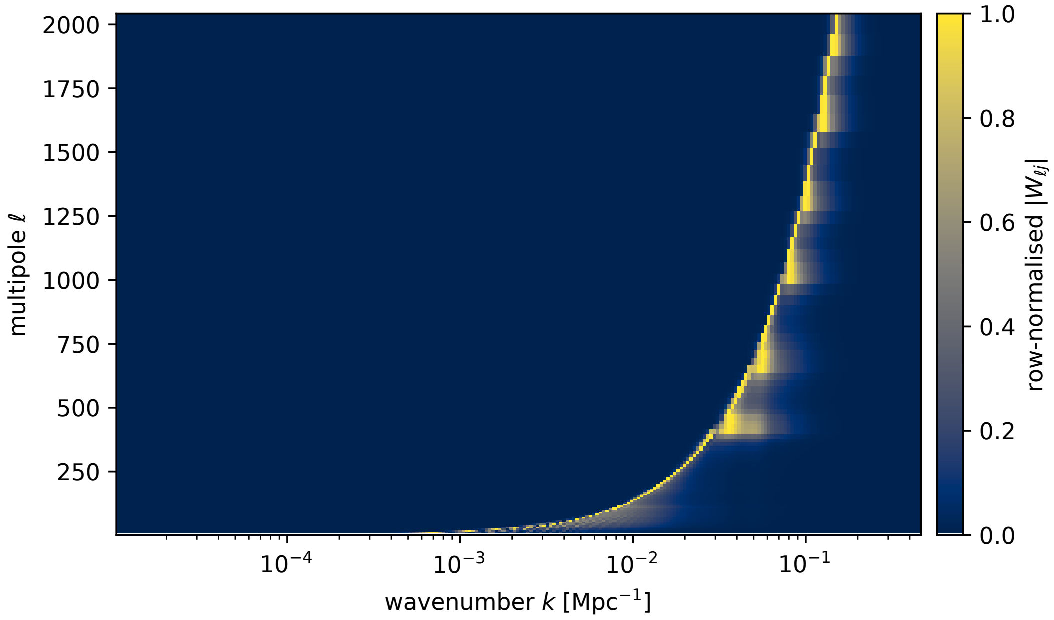

Figure 1. The discretised synthetic CLASS operator. Each row maps the primordial spectrum on the $\ln k$ grid to one temperature datum; rows are normalised for visibility. The broad response and its changing support are the source of the inverse problem.

For the numerical calculation we set $u = \ln k$ and work with the order-unity field

$$f(u) = \frac{\mathcal{P}_{\mathcal{R}}(e^u)}{A_0}, \qquad A_0 = 2.1 \times 10^{-9}. \tag{2.4}$$

Quadrature and, for Planck, the bandpower windows turn the forward map into

$$\boldsymbol{y} = \boldsymbol{W}\boldsymbol{f} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}). \tag{2.5}$$

The matrix $\boldsymbol{W}$ includes A_0 and therefore predicts the data in their physical units. We keep the dimensionless spectrum and the $\mathrm{d}\ln k$ measure explicit throughout.

2.2 Gaussian-process inversion

We assign f a Gaussian-process prior with mean m and radial-basis-function kernel

$$K(u, u') = \sigma_f^2 \exp\left[-\frac{(u-u')^2}{2\gamma^2}\right]. \tag{2.6}$$

Our reference choice in both calculations is $m = 1$, $\sigma_f = 1$ and $\gamma = 1.5$. The mean is a scale-invariant spectrum, not the tilted power law that we later reconstruct. The prior permits smooth departures over order-unity intervals in $\ln k$. We keep $m = 1$ fixed and vary the two RBF hyperparameters in table 2. A feature analysis should instead infer or marginalise over them.

Because equations (2.5) and (2.6) are Gaussian and linear, the posterior is analytic [12, 13]:

$$\overline{\boldsymbol{f}} = \boldsymbol{m} + \boldsymbol{K}\boldsymbol{W}^{\mathsf{T}}(\boldsymbol{W}\boldsymbol{K}\boldsymbol{W}^{\mathsf{T}} + \boldsymbol{\Sigma})^{-1}(\boldsymbol{y} - \boldsymbol{W}\boldsymbol{m}), \tag{2.7}$$

$$\boldsymbol{K}_{\text{post}} = \boldsymbol{K} - \boldsymbol{K}\boldsymbol{W}^{\mathsf{T}}(\boldsymbol{W}\boldsymbol{K}\boldsymbol{W}^{\mathsf{T}} + \boldsymbol{\Sigma})^{-1}\boldsymbol{W}\boldsymbol{K}. \tag{2.8}$$

The posterior returns to the prior where $\boldsymbol{W}$ has little sensitivity. That behaviour is part of the answer: an apparently smooth extrapolation outside the supported range must not be mistaken for a measurement.

FrediPy provides the Gaussian process, covariance kernel, integral operator and linear noisy constraint that evaluate these expressions. Our adapter passes the precomputed cosmological quadrature to its public `Integral` and `LinearEquality` classes. Since the FrediPy process is zero-mean, the adapter conditions on $\boldsymbol{y}-\boldsymbol{W}\boldsymbol{m}$ and adds $\boldsymbol{m}$ back afterwards. It passes `cov_y=` $\boldsymbol{\Sigma}$ as a covariance—not a vector of standard deviations. The repository tests the returned mean and covariance against a separate dense Gaussian evaluation of equations (2.7)–(2.8).

3 Controlled reconstruction from CLASS

The first test should answer one question without observational complications: if the input spectrum is known, does the full chain recover it? We generate an unlensed scalar TT spectrum with CLASS 3.3.4 using

$$A_s = 2.1 \times 10^{-9}, \qquad n_s = 0.9649, \qquad k_* = 0.05\,\mathrm{Mpc}^{-1}. \tag{3.1}$$

We use flat scalar adiabatic initial conditions and fix $h = 0.6736$, $\omega_b = 0.02237$, $\omega_{\mathrm{cdm}} = 0.1200$, $\tau_{\mathrm{reio}} = 0.0544$, $N_{\mathrm{ncdm}} = 0$ and $N_{\mathrm{ur}} = 3.044$. The density parameters are Planck-like [1], but the absence of a massive-neutrino species is part of this proof-of-concept setup. We sample 125 multipoles between $\ell = 2$ and 2000 and reconstruct on 240 logarithmic nodes covering $1.05 \times 10^{-5} < k < 0.455\,\mathrm{Mpc}^{-1}$.

For this synthetic test we convert `CLASS.raw_cl` to the conventional rescaled spectrum

$$D_\ell \equiv \frac{\ell(\ell+1)}{2\pi} C_\ell. \tag{3.2}$$

The deterministic D_ℓ values form the target vector; no random noise realisation is added. We assign a diagonal covariance with standard deviations $\sigma_\ell = 0.01 D_\ell$, or variances $(0.01 D_\ell)^2$, to define a simple conditioning problem. This covariance is a numerical convention for this closure test, not a model of cosmic variance or an experiment. The operator is constructed independently from the CLASS line-of-sight sources and a finer k quadrature. Its direct forward spectrum agrees with `CLASS.raw_cl` to a maximum fractional difference of 9.42×10^{-4} across the sampled multipoles.

Figure 2 shows the result. On the fixed interval

$$3 \times 10^{-4} < k < 0.15\,\mathrm{Mpc}^{-1}, \tag{3.3}$$

the maximum absolute fractional error of the posterior mean is 7.46×10^{-4} and the root-mean-square error is 4.16×10^{-4}. A constant spectrum at the prior mean differs from the truth by as much as 16.3% on the same interval, so the agreement is not a trivial restatement of the prior. Across the complete k grid, by contrast, the maximum error rises to about 19% as the reconstruction loses support and returns to $m = 1$. Quoting the restricted range is therefore essential.

This is a closure test, not an independent test of CLASS. The target and the transfer sources come from the same Boltzmann code, although the direct angular spectrum and the reconstructed operator use separate numerical paths. What it does establish is the

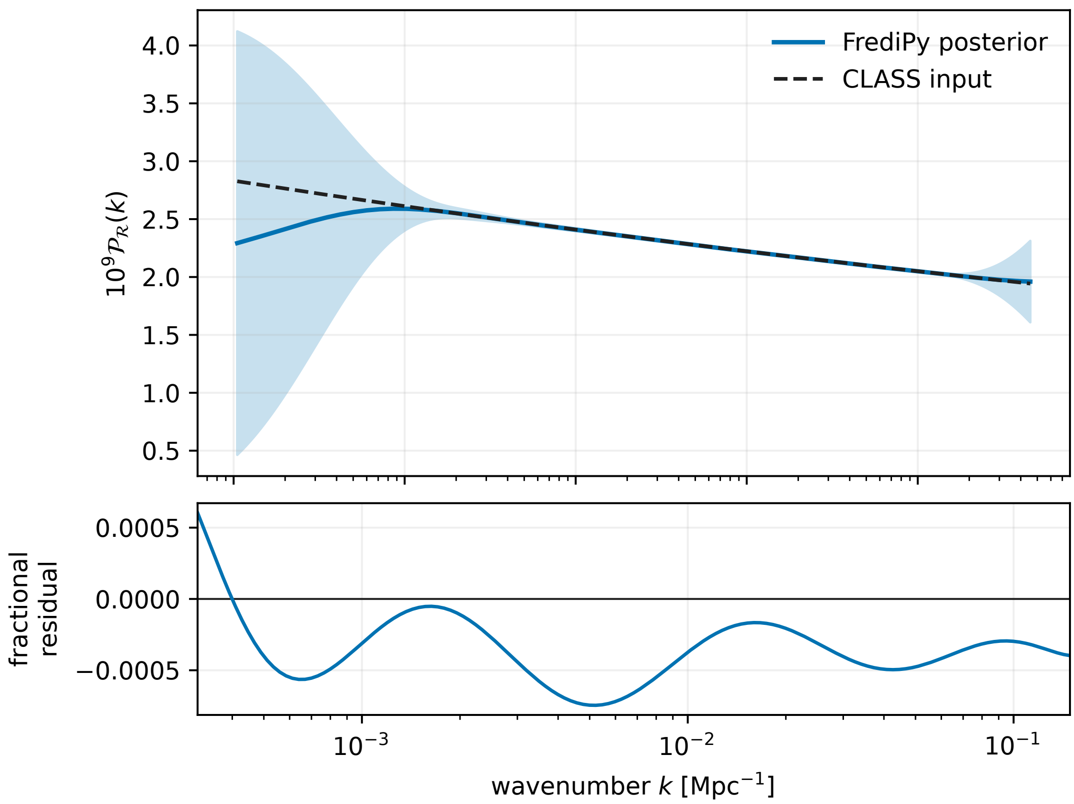

Figure 2. Controlled CLASS reconstruction. The upper panel compares the injected primordial power law with the FrediPy posterior mean and its ± 1 pointwise posterior standard deviation; this is not a simultaneous band. The lower panel shows the fractional reconstruction residual on the fixed evaluation interval. Outside this interval, visible in the upper panel, the result eventually reverts towards its constant prior mean.

desired proof of concept: with the correct dimensionless-spectrum normalisation and sufficient quadrature resolution, FrediPy recovers the primordial power law that generated the synthetic CMB.

4 A conditional reconstruction from Planck TT

We next replace the synthetic vector by the first 215 TT bandpowers of the Planck PR3 `plik_lite` high-ℓ likelihood, covering $30 \leq \ell \leq 2508$ [14, 15]. We retain the full 215×215 bandpower covariance. The exact published binning windows were applied upstream and are encoded in the compact bandpower operator $\boldsymbol{W}$. The background parameters remain fixed to the values used in the synthetic test. The forward model consists of the unlensed linear response of equation (2.2) plus a fiducial lensed-minus-unlensed bandpower template. Calibration is fixed to unity. Thus the dependence on $\mathcal{P}_\mathcal{R}$ remains linear, at the cost of holding the lensing correction fixed.

Table 1 summarises what changes between the two calculations. The same grid, prior and public FrediPy conditioning are used in both.

To identify the range to which this discretised operator is directly sensitive, we whiten its rows using the data covariance. If $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^\mathsf{T}$, the noise-weighted sensitivity score of primordial

Table 1. The two calculations retained in the streamlined analysis.

	Synthetic closure	Planck illustration
Temperature data	CLASS-derived D_ℓ, 125 values	PR3 high-ℓ TT, 215 bins
Covariance	diagonal, $\sigma_\ell = 0.01 D_\ell$	full `plik_lite` covariance
Lensing	absent	fixed additive template
Calibration	not applicable	fixed to unity
Cosmology	fixed	fixed
GP hyperparameters	fixed RBF	fixed RBF; sensitivity scan
Purpose	recovery of known truth	conditional adequacy check

node j is

$$s_j = \left\| \boldsymbol{L}^{-1} \boldsymbol{W}_{:j} \right\|_2. \tag{4.1}$$

We report comparisons where s_j is at least 1% of its maximum. This selects 89 contiguous nodes over the interval, rounded to three significant figures,

$$k \in [0.00399, 0.204]\,\mathrm{Mpc}^{-1}. \tag{4.2}$$

The threshold is an explicit reporting convention rather than a confidence boundary.

For comparison with the flexible posterior, we fit the two parameters of equation (1.1) at $k_* = 0.05\,\mathrm{Mpc}^{-1}$ using the same operator, lensing template and covariance. We minimise

$$Q = (\boldsymbol{d} - \boldsymbol{t})^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} (\boldsymbol{d} - \boldsymbol{t}). \tag{4.3}$$

Here $\boldsymbol{d}$ is the observed bandpower vector and $\boldsymbol{t}$ is the prediction. The optimum is

$$A_s = 2.094 \times 10^{-9}, \qquad n_s = 0.961, \tag{4.4}$$

with $Q = 218.90$. Evaluating the FrediPy posterior mean in the same forward model gives $Q = 218.51$. For reference, the unoptimised fiducial power law $(A_s, n_s) = (2.1 \times 10^{-9}, 0.9649)$ gives $Q = 225.37$.

Within the sensitivity range, the posterior mean and best-fitting power law differ by at most 1.24%, with a root-mean-square difference of 0.334%. Their largest pointwise separation is 0.48 posterior standard deviations, and every retained node lies within one pointwise standard deviation. As figure 3 shows, the non-parametric reconstruction is therefore pointwise close to a straight line in $\ln \mathcal{P}_\mathcal{R}$ versus $\ln k$. The improvement $\Delta Q = 0.39$ of the flexible posterior mean is small; moreover, Q alone does not penalise the freedom of a function. We consequently do not interpret it as evidence for primordial features.

Within this small scan, the qualitative Planck result is not peculiar to the reference hyperparameters. Halving or doubling either RBF hyperparameter keeps the maximum separation from the fitted power law below 2.9% and ΔQ at or below 1.80, as table 2 shows. A deliberately shorter correlation length, $\gamma = 0.5$, permits a 5.38% separation and gives $\Delta Q = 3.45$. This is the expected direction: reducing γ admits finer structure. The scan keeps the cosmology, lensing template, calibration, prior mean and reporting range fixed. It is not a substitute for hyperparameter marginalisation or a global feature test.

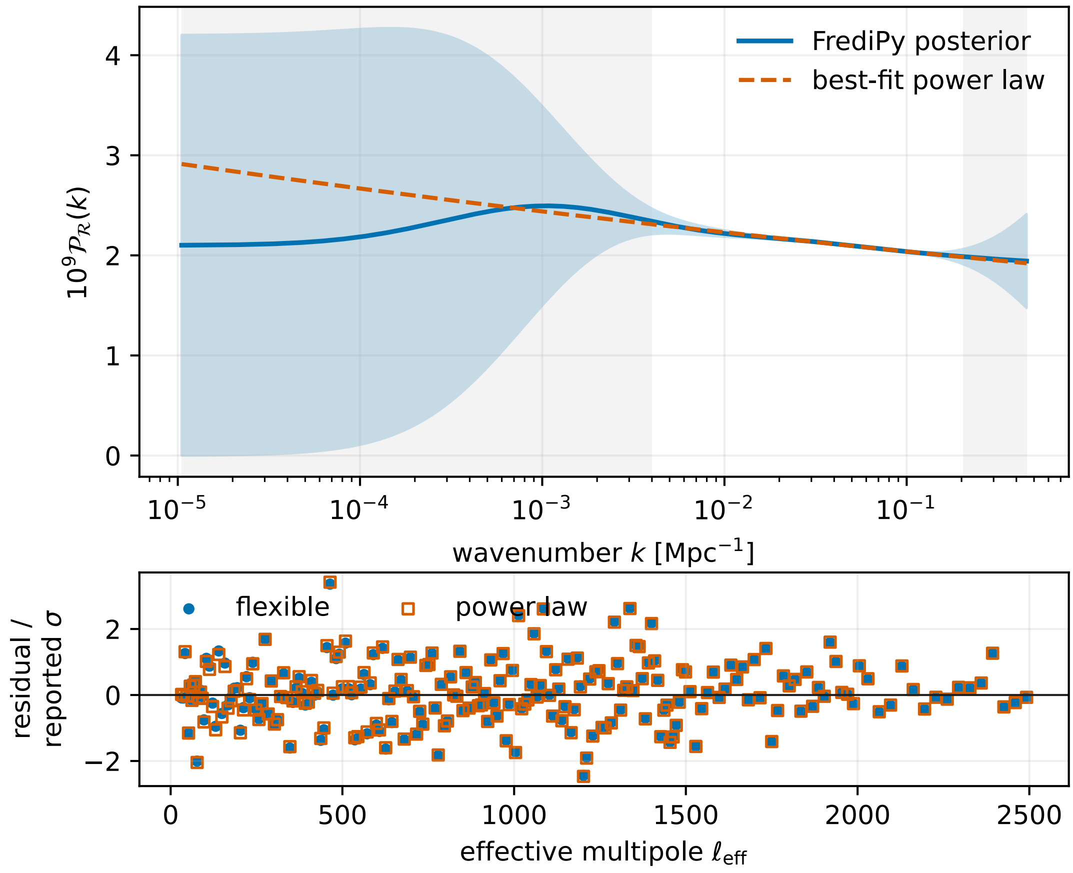

Figure 3. Conditional Planck PR3 reconstruction. The FrediPy posterior mean and its ±1 pointwise posterior standard deviation are compared with the best-fitting power law of equation (4.4); the band is not simultaneous, and grey regions lie outside the direct-sensitivity range. The lower panel shows the bandpower residuals divided by the reported diagonal errors. Their correlations are retained in the fits and in the quoted Q values.

Table 2. Hyperparameter sensitivity of the conditional Planck reconstruction. The maximum fractional difference between the flexible posterior mean and the same best-fitting power law is evaluated over the 89-node sensitivity range. All other assumptions are fixed. The difference $\Delta Q = Q_{\rm PL} - Q_{\rm flexible}$ is descriptive; it is neither a likelihood-ratio statistic nor a feature-significance measure.

σ_f	γ	Maximum difference	ΔQ
1.0	1.50	1.24%	0.39
0.5	1.50	0.63%	0.35
2.0	1.50	1.48%	0.41
1.0	0.75	2.88%	1.80
1.0	3.00	0.39%	0.22
1.0	0.50	5.38%	3.45

5 Scope of the result

The synthetic result is the central claim of this paper: FrediPy correctly solves the CLASS primordial-spectrum inverse problem in a controlled setting. The Planck calculation has a narrower role. It shows that real bandpowers and a full covariance can be supplied to the same machinery, and that the output does not demand visible departures from a power law under the stated conditions.

Several ingredients would have to change before turning the illustration into a parameter constraint or feature search. The background cosmology, calibration and lensing response should be varied jointly with $\mathcal{P}_{\mathcal{R}}$; low-$\ell$ temperature and polarisation information should be included; foreground and likelihood approximations should be treated at the level appropriate to the experiment; and the kernel family should be varied. The small sensitivity scan above does not replace hyperparameter marginalisation. Pointwise posterior bands are correlated and do not define a global feature test. The Gaussian prior also does not enforce positivity, although the posterior means shown here remain positive.

A fixed smoothness prior can suppress narrow features, while a more permissive prior can fit fluctuations without making them significant. Individual wiggles in figure 3 therefore have no direct physical interpretation. A feature analysis would instead embed the reconstruction in a joint likelihood and compare prior choices on simulated skies with known coverage.

6 Conclusions

The primordial curvature spectrum provides a direct cosmological example of a Fredholm equation. At fixed cosmology, CLASS supplies the kernel, while FrediPy supplies the Gaussian-process conditioning. The resulting calculation is small enough to inspect: a matrix forward model, a data covariance and the closed-form posterior of equations (2.7) and (2.8).

The controlled test succeeds. A tilted primordial power law is recovered from synthetic CLASS TT data with a maximum fractional error below 7.5×10^{-4} on the fixed interval. Applying the same interface to Planck high-ℓ TT gives a posterior mean close to a fitted power law over the range with direct operator sensitivity. In this fixed setup, a featureless power law explains the bandpowers about as well as the flexible posterior mean. Changing the RBF hyperparameters alters the visible departures, as expected, but the small scan does not turn them into a feature-significance result.

This example shows that FrediPy can perform the primordial-spectrum reconstruction and provides a reproducible starting point for a later joint CMB analysis. It does not claim that the data exclude features or prefer a particular inflationary model.

Data and code availability

The accompanying repository contains the analysis code, focused tests, paper source, figures and compact numerical inputs needed to reproduce every number quoted here. It recomputes the inversion through FrediPy from checksum-verified, precomputed operators rather than loading a cached posterior. Raw CLASS and Planck operator generation remains provenance-linked to the larger validation repository. The software is released under the MIT licence; the Planck-derived input retains its separate upstream terms. FrediPy 0.2.1 and CLASS 3.3.4 are cited explicitly [9, 11].

Acknowledgments

I acknowledge support from the John and Pat Hume Scholarship, the Swiss Study Foundation and the Friedrich Naumann Foundation for Freedom. OpenAI Codex (GPT-5), including delegated Codex agents, was used in August 2026 for repository restructuring, code development, numerical analysis and cross-checks, and manuscript drafting and editing. I remain responsible for the scientific content, code and final manuscript.

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